Optima Academy Online

An Education Experience Company

Calculus Honors · Teacher Manual

Quarter 1: Limits, Continuity, and the Derivative · Florida Course 1202300

PURPOSE OF THIS MANUAL

This manual equips a facilitator to run Quarter 1 of Calculus Honors as an asynchronous Canvas course that a live class can also use unchanged. It covers the teaching philosophy, the weekly rhythm, the assessment architecture and your time budget, module-by-module teaching notes with standards alignment, answer-key locations, and the technical notes needed to deploy and maintain the course. Standards codes appear here for your reference; they never appear on student-facing pages.

TEACHING PHILOSOPHY

Calculus at Optima is taught as one of the liberal arts: rigorous, historically grounded, and beautiful. Every lesson integrates procedure with concept. When a technique is taught, students also learn why it works; when a concept is introduced, students get enough worked practice to actually do it. Insist on clean, vertical written work with aligned relations and labeled answers, and use the dialogue prompts built into each lesson so students reason aloud and in writing. Brief historical vignettes (Newton and Leibniz, Cauchy and Weierstrass, Archimedes) situate each idea in the human story of the subject; keep them short and let the mathematics lead.

THE WEEKLY RHYTHM

- **Two lessons per week.** Interactive pages with embedded worked examples, self-check widgets, and a central visual. Ungraded; students work the practice in a notebook.
- **One problem set per week.** A printable worksheet worked by hand and uploaded. Scored on completion and good-faith effort.
- **One auto-graded quiz per week.** Eight questions, immediate feedback, no teacher cost.
- **Per module (two weeks):** a module map, an intro survey, a discussion, a Module Challenge (your one deep-feedback task), a VR activity, and an outro survey.

ASSESSMENT ARCHITECTURE AND YOUR TIME BUDGET

The course is built to the 500:1 ratio. Design intent per module holds roughly to 60% student-and-material, 25% student-and-community, 15% student-and-teacher.

Tier	Items	Teacher time
1 · Material (auto)	2 weekly quizzes, 2 weekly problem sets (completion), in-lesson widgets, VR (completion)	~0 hrs
2 · Community	1 discussion, outro survey triage	~2 to 3 hrs/wk
3 · Teacher	1 Module Challenge (four-criterion rubric)	~8 to 12 hrs/module cycle
4 · Scaffold	module map, intro and outro surveys	~15 to 20 min/module (survey analytics)

> Mine the intro and outro survey analytics: the confidence Likert item is your early-warning signal, and the "what remains unclear" field tells you where to drop public comments in the discussion.

DEPLOYMENT NOTES

- **GitHub-hosted, embedded in Canvas:** interactive lessons, module maps, course home, syllabus, teacher manual, rubrics. These are full HTML documents. Host on GitHub Pages and embed via iframe.
- **Canvas-native (build directly, no custom code):** quizzes (New Quizzes), the intro/outro survey wrappers (Canvas Quiz in survey mode), assignment submission points, and discussions. The `.md` spec files give the exact questions and structure.
- **Problem sets and Module Challenges:** distribute the generated PDF worksheet; collect as a Canvas Assignment file upload.
- **Math rendering:** interactive pages load MathJax from a CDN in the page head, because Canvas's built-in MathJax does not reach content inside an iframe. Canvas-native items use Canvas's own MathJax via `\( \)` delimiters.

> **Sequencing note (important):** the scope and sequence lists L'Hôpital's Rule (MA.912.C.1.8) in Unit 1. L'Hôpital requires the derivative, so it cannot be taught before Unit 2. Quarter 1 introduces indeterminate forms conceptually and names L'Hôpital as the tool to be proved and used in early Quarter 2. Do not attempt to teach the rule itself in Module 2.

MODULE-BY-MODULE TEACHING NOTES

Module 1: The Idea of a Limit (Weeks 1 to 2) · Virtue: Wonder

Standards: MA.912.C.1.1, C.1.2, C.1.3, C.1.4. **Essential question:** How can we speak precisely about what a function approaches, even where it is never defined?

Lessons: 1.01 what a limit is (tables and graphs); 1.02 the limit laws and direct substitution; 1.03 the 0/0 form (factor and rationalize); 1.04 one-sided limits and existence. **Watch for:** students conflating the limit with the function value at the point. The whole quarter depends on keeping those separate.

Module 2: Limits Without Bound and Special Limits (Weeks 3 to 4) · Virtue: Perseverance

Standards: MA.912.C.1.5, C.1.6, C.1.7 (C.1.8 conceptual only). **Essential question:** What happens to a function at the edges of its domain and where it grows without bound?

Lessons: 2.01 limits at infinity and horizontal asymptotes (slider visualizer); 2.02 infinite limits and vertical asymptotes; 2.03 the Squeeze Theorem and $\frac{\sin x}{x}$; 2.04 indeterminate forms (name L'Hôpital, defer the rule). **Watch for:** students writing "the limit equals infinity" as if infinity were a number; frame it as a description of unbounded behavior.

Module 3: Continuity and the Great Theorems (Weeks 5 to 6) · Virtue: Honesty

Standards: MA.912.C.1.9, C.1.10, C.1.11. **Essential question:** What does it mean for a function to have no gaps, and what can we prove because of it?

Lessons: 3.01 the three-part definition of continuity; 3.02 classifying discontinuities; 3.03 the Intermediate Value Theorem (root existence); 3.04 the Extreme Value Theorem and Unit 1 synthesis. **Watch for:** students treating IVT and EVT as procedures rather than existence guarantees; stress the role of the hypotheses (closed interval, continuity).

Module 4: The Derivative, Definition and Meaning (Weeks 7 to 8) · Virtue: Self-government

Standards: MA.912.C.2.1, C.2.2, C.2.9 (with C.3.8 rate-of-change introduced). **Essential question:** How do we capture the exact rate of change at a single instant?

Lessons: 4.01 average vs instantaneous rate (secant-to-tangent slider); 4.02 the derivative at a point and the tangent line; 4.03 the derivative as a function; 4.04 differentiability and continuity. **Watch for:** algebra slips inside the difference quotient; require the full expansion in worked solutions.

ANSWER KEYS AND GRADING

Every problem set and Module Challenge has a teacher-only key in that module's `answer_keys/` folder. Problem sets are scored on completion; use the keys for spot checks and for the discussion and Module Challenge. Grade Module Challenges with the shared four-criterion rubric (published in the Rubrics document): Understanding, Quality of Reasoning, Clarity and Precision, and Intellectual Honesty. Keep the rubric identical across modules so five-minute-per-student grading stays feasible.

VR FACILITATOR NOTES

Each module has one solo VR activity that feeds the module's central idea (M1 Walk the Limit; M2 The Horizon That Is Never Reached; M3 Continuous or Broken; M4 From Secant to Tangent). Each is one in-headset session of about 20 to 25 minutes, submitted through Canvas as a snapshot plus a short spatial recording or written reflection. The activity documents include a facilitator note describing common sticking points and which IFX and tools to be ready to demonstrate. Support students around the activity; do not deliver a separate live version.